

ColdCallAI — Full Task List

AI-Powered Outbound Calling System — Complete Build Task Register

5 Build Phases	58 Total Tasks	60–70h Total Effort incl. Connection	0 BizFinder Code Changes	100 Calls Per Learning Batch
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Phase 1 — Voice Agent (Browser)			12 tasks · 10–14h · Gate: P50 latency < 700ms · 10 test scenarios pass
1	T01	InWorld STT streaming provider class	☐
2	T02	InWorld TTS Realtime streaming provider class	☐
3	T03	Latency spike test — measure full round-trip, gate at < 1200ms	☐
4	T04	Voice variants: load 4–6 InWorld voices, name them, config per segment	☐
5	T05	Real-time loop: VAD → STT → Qwen (streaming) → TTS → audio out	☐
6	T06	Barge-in: VAD detects speech during TTS → flush queue in < 100ms	☐
7	T07	Silero VAD integration with exact parameters (20ms frames, 350ms end-of-turn)	☐
8	T08	In-call memory: CallMemory class with sliding window + confirmed_facts	☐
9	T09	Sales script framework: system prompt template + state machine	☐
10	T10	Graceful close: detect hangup, collect outcome, emit call_ended	☐
11	T11	Browser WebRTC harness (FastAPI WebSocket ↔ browser mic/speaker)	☐
12	T12	Live end-to-end test: runs all 10 test scenarios (objections, barge-in, close)	☐

Phase 2 — Orchestrator + Dashboard			18 tasks · 12–16h · Gate: 50-lead CSV runs in UI · analytics + recording playback functional
1	T43	DB schema migrations (all tables: leads, calls, campaigns, variants, qa_reports)	☐
2	T44	CSV import: parse, validate, E.164 normalize, bulk insert, embed to Qdrant	☐
3	T45	Phone normalization utility (E.164 + line type lookup)	☐
4	T49	Flexible JSONB curiosity columns on lead table	☐
5	T50	PostgreSQL + Qdrant setup (connection pooling via PgBouncer)	☐
6	T21	Lead scoring: Qwen 0–100 qualification prompt + batch scoring on import	☐
7	T22	Talking points: industry-level personalization generator	☐
8	T23	Orchestrator: LeadBrief builder with Thompson Sampling voice/script selection	☐
9	T24	Variant tagging: record exact voice/script/opener per call in call_result	☐
10	T25	QA ↔ orchestrator contract: read qa_report, apply next_batch_recommendations	☐
11	T26	Transcript generation: post-call STT + speaker diarization [AGENT]/[PROSPECT]	☐
12	T27	Outcome classifier: Qwen judge → outcome enum, sentiment, objections list	☐
13	T28	Structured answer extraction: JSON output from Qwen judge	☐
14	T29	Cost capture: track minutes + tokens + TTS characters per call	☐
15	T51	Campaign UI: create, configure, CSV upload, start/pause/stop controls	☐
16	T52	Live run monitor: real-time progress, active call view, kill switch	☐
17	T53	Results analytics: conversion charts by voice/script/industry/geo/time	☐
18	T55	Recording playback: audio player + synchronized transcript view	☐

Phase 3 — Live Calling (FreePBX)			12 tasks · 10–14h · Gate: 10 real calls · AMD > 70% · legal checklist passed
1	T13	FreePBX ARI setup + AudioSocket TCP server (port 9092/9093)	☐
2	T14	Outbound originate: AMI originate from campaign queue	☐
3	T15	Caller-ID pool: selection logic (geographic match, usage rotation, flag detection)	☐
4	T16	Call recording: FreePBX MixMonitor + post-call file sync	☐
5	T17	AMD integration: amd.conf tuning + AMDSTATUS branch in dialplan	☐
6	T18	Concurrency control: max parallel lines + rate caps + retry queue	☐
7	T19	Calling window enforcement: timezone-aware, reject outside 8am–9pm local	☐
8	T20	DNC suppression: internal DB + check before every dial + opt-out handler	☐
9	T30	After-hours dumb script: 2 questions, classify response, hang up	☐
10	T31	Response classifier: live/voicemail/answering-service/rings-to-answer	☐
11	T32	Probe dataset: schema + export for analysis	☐
12	T33	Stats report: after-hours probe analytics	☐

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Phases 4 & 5 · Deploy & Ops

Phase 4 — BizFinder AI Connection			3 tasks · 8–12h · Gate: lead pulled → call placed → outcome written back · zero BizFinder schema changes
1	T46	BizFinder PostgreSQL read API: pull qualified leads into ColdCallAI campaign queue	☐
2	T47	Business-info replication: copy lead data to local DB for QA context	☐
3	T48	Outcome write-back: post call result, disposition, and recording link to BizFinder record	☐

Phase 5 — Self-Improving Loop			10 tasks · 12–16h · Gate: QA batch auto-runs · variant approved in dashboard · trend visible across 3+ batches
1	T34	Voice QA hookup: intake/output contract, POST /qa/run-batch endpoint	☐
2	T35	Voice performance ranking: QA ranks voices by segment conversion rate	☐
3	T36	Batch runner: auto-trigger at 100 calls, batch status tracking	☐
4	T37	Aggregation: per-batch rollup + per-variable breakdown	☐
5	T38	Bayesian models: Beta-Binomial per variant + Thompson Sampling allocator	☐
6	T39	Correlation analysis: feature matrix → Pearson/Cramér's V vs conversion	☐
7	T40	Promote/demote: threshold rules + update variant statuses	☐
8	T41	Variant generator: Qwen proposes new openers from best-performing transcripts	☐
9	T42	Human review gate: pending proposals UI — approve / edit / reject	☐
10	T54	Loop view: batch timeline, variant lineage tree, voice leaderboard	☐

Ongoing — Deploy & Ops			3 tasks · runs across all phases · Gate: production deploy stable · secrets never committed
1	T56	Coolify service deploy: Dockerfile, docker-compose, environment variables	☐
2	T57	Secrets management: .env per environment, never commit credentials to git	☐
3	T58	Cost monitoring: daily/monthly caps, 80% alert threshold	☐

Task Count by Phase

Ph.1 Voice — 12 tasks

Ph.2 Dashboard — 18 tasks

Ph.3 FreePBX — 12 tasks

Ph.4 BizFinder — 3 tasks

Ph.5 Loop — 10 tasks

Ops — 3 tasks

Total — 58 tasks